

Assessing hierarchical categories of evidentiality in Chinese in terms of readers' certainty levels

[從讀者角度檢測中文實據性類別之間的排序]

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This study examines whether knowledge encoded in evidential categories is perceived differently by readers with varying levels of certainty. A questionnaire with a five-point scale was conducted to collect readers' judgments about events introduced by evidential markers in Chinese. The evidential markers were manually categorized into categories, and inter-rater agreement was calculated. The findings reveal two hierarchies in terms of readers' degree of certainty in judging the veracity of the knowledge: visual > hearsay and inference > quotative > hearsay. Additionally, three factors were observed to influence readers' judgments about events: (a) whether a detailed event depiction is provided, (b) whether knowledge of an event is provided by more than one source, and (c) whether knowledge of an event has a close relationship with readers' political standpoints or personal societal beliefs.

Keywords: evidentiality, evidential category, readers' certainty levels, encoded knowledge, evidential hierarchy

關鍵字：示證性、示證式類別、讀者的確定程度、知識編碼、示證式階層

1. Introduction

Events occur daily and are widely reported in the media. An event is often expressed using evidentiality, which conveys not only knowledge of the event but also the speaker's degree of certainty about its occurrence (Saurí 2008). For example, when a news report states, "Mary assures/conjectures that John has submitted his proposal," the speaker may express a higher or lower degree of certainty regarding John's submission through the use of evidential markers (e.g., *assures* and *conjectures*).

Evidentiality is primarily recognized as encoding knowledge or the source of information (Aikhenvald 2004). In many languages, evidentiality is expressed grammatically, while some languages (e.g., Chinese) express it through evidential-like expressions (i.e., the evidentiality strategy, Aikhenvald 2004). Various perspectives have been proposed regarding the definition of evidentiality and its relationship to the speaker's degree of certainty about information (epistemic modality) (e.g., Chafe 1986, Palmer 1986, Dendale & Tasmowski 2001). For example, Aikhenvald (2004) claimed that evidentiality is an independent system marking the source of information, without involving the speaker's degree of certainty in the definition of evidentiality. More broadly, Chafe (1986) considered evidentiality to encode not only the source of information but also the speaker's attitude toward the knowledge. In this study, the broader sense of evidentiality is adopted to explore the speaker's attitude expressed through knowledge that they intend for readers to interpret (Weber 1986).

Extensive scholarly discussion has addressed the categorization of encoded knowledge in evidentiality. Subsequent research has investigated whether these evidential categories reflect different degrees of speaker certainty in discourse. As an illustration, Sweetser (1990) demonstrated that visual (e.g., *see*) and sensory (e.g., *smell*) evidentials generally convey a high degree of certainty. Similarly, Saurí (2008) indicated that visual evidentials may be considered more certain than inferential evidentials (e.g., *deduce*). Various hierarchical rankings of evidential categories based on their certainty levels have been proposed. For example, Willett (1988) proposed that direct evidentials (e.g., visual, auditory, other sensory) present a higher degree of certainty than indirect and hearsay evidentials, with hearsay evidentials having the lowest certainty levels. In Chinese, Ma (2011) categorizes evidentials into five categories (i.e., visual, auditory, inference, quotative, hearsay) and argued that the degree of certainty in these evidential categories is distributed in descending order. However, these rankings have been primarily discussed in theoretical terms, with limited empirical assessment of whether readers actually perceive the knowledge encoded in evidential categories in the same hierarchical order.

This study aims to investigate whether the hierarchy of Chinese evidential categories proposed by Ma (2011) is consistently reflected in readers' assessment of event knowledge as expressed by evidentials. Additionally, we will explore factors influencing readers' degrees of certainty regarding perceived knowledge of events. We utilized a dataset of Chinese news sentences from a previous study (Chang & Hsieh 2020) and employed a five-point scale to measure readers' degrees of certainty regarding the various types of knowledge encoded in the hierarchical evidential categories.

2. Literature review

2.1 The hierarchical categorization of evidentiality

Several studies have proposed categorizing evidential markers into hierarchies in terms of the degree of certainty. While typologists have proposed various hierarchical evidential categories across languages (Willett 1988, Lesage et al. 2015), many studies have considered the influence of cultural or societal factors on each language's evidential system and focused on constructing hierarchies tailored to specific languages (Aikhenvald 2014, Bernardez 2017, San Roque 2019). For example, Barnes (1984) explored the hierarchy of evidentiality in the Tuyuca language, arranged as visual > non-visual > apparent > second-hand > assumed, with visual evidentials providing the highest degree of certainty. Oswalt (1986) proposed a hierarchy of evidentials in the Kashaya language: performative > factual > visual > auditory > inferential > quotative. In this language, visual evidentials received relatively lower certainty compared to Tuyuca, demonstrating that evidential hierarchy is culturally sensitive.

In Chinese, several studies have explored the categorization of evidentials (e.g., Niu 2005, Fang 2006, Zhu 2006), and the majority have emphasized categorizing evidential markers into various groups rather than establishing a hierarchical ranking based on the degree of certainty. However, Ma (2011) stands out as the first and only study to establish a hierarchy of Chinese evidentiality, consisting of five categories representing varying degrees of certainty: visual > auditory > inference > reported with a known source (hereafter quotative) > reported with an unknown source (hereafter hearsay). This hierarchy was developed by analyzing how speakers convey their attitudes through evidentials, based on observations of Chinese natural language. While the hierarchy is tailored to analyzing certainty levels in Chinese evidentials, it also shows that direct evidentials (e.g., visual, auditory) convey higher certainty than indirect evidentials (e.g., inference, quotative, hearsay), generally coinciding with the findings of most studies (e.g.,

Willett 1988, Saurí 2008, Martínez Vera 2024). Furthermore, even though the hierarchy was developed based on natural language, Ma (2011) stated that it requires further examination with empirical evidence. Therefore, this study adopts Ma's hierarchical scheme to investigate how readers judge the knowledge conveyed by these evidential categories. Below are the five evidential categories with their corresponding interpretations, as outlined in the work of Ma (2011).

- (1) a. Visual: the evidential involving knowledge obtained through visual observation.
- b. Auditory: the evidential specifying knowledge obtained through hearing.
- c. Inference: the evidential conveying knowledge based on explicit given evidence.
- d. Quotative: the evidential presenting knowledge conveyed by an explicit source provider of an event.
- e. Hearsay: the evidential showing knowledge presented without knowing its source provider of an event.

2.2 The scale of certainty levels

For collecting readers' degrees of certainty regarding knowledge of an event, de Marneffe, Manning & Potts (2012) utilized a seven-point Likert-type scale inspired by Saurí (2008). The seven-point scale incorporated "certainly happened," "probably happened," "possibly happened," "certainly did not happen," "probably not did happen," "possibly did not happen," and "unknown." This scale was further adopted by Chang & Hsieh (2020), who compiled a Chinese dataset collected with readers' degrees of certainty following the same questionnaire design as depicted in de Marneffe, Manning & Potts (2012). While the seven-point scale provided a fine-grained assessment of reader's certainty levels, feedback from participants in Chang & Hsieh (2020) indicated ambiguity in interpreting certain probability levels (e.g., "probably happened"). These probability levels could reflect scenarios where readers have slightly lower certainty, but they might also represent cases where readers have slightly higher certainty. In other words, the criteria for assigning a probability judgment to an event are less clear and more ambiguous compared to the criteria for "certainly happened" and "possibly happened" levels. This situation is also highlighted by Su & Liu (2012), who reported that individuals often struggle to reach a consensus in such cases. Thus, to avoid ambiguity, there is a need to adjust the number of scale points for collecting degree of certainty levels.

While studies have proposed various numbers of scale points for measuring people's judgments in questionnaires, Grilli & Rampichini (2012) concluded that a five-point scale is typical as it produces an acceptable bias with a minimum

number of scale points. Additionally, Garratt, Helgeland & Gulbrandsen (2011) conducted an experiment and concluded that the five-point scale results in a less skewed distribution and lower ceiling and floor effects compared to scales with more points. Building upon previous literature, we have decided to adopt a five-point scale in this study. Details of the five-point scale are elaborated in Section 3.2.

3. Methodology

3.1 Annotation of evidential categories

The dataset utilized in this study was compiled by Chang & Hsieh (2020), from which a total of 673 Chinese news sentences containing events introduced by evidential markers were selected for analysis. An illustrative example of a sentence tagged with an evidential marker (e.g., 表示 *biǎoshì* 'say') and a predicate encoding event knowledge (e.g., 決定 *juéding* 'determine') in the dataset is provided in (2).¹

- (2) 專家表示，地震的規模和深度將決定其對當地社區的影響。

Zhuānjiā **biǎoshì**, dìzhèn de guīmó hàn shēndù jiāng juéding qí duì dāngdì shèqū de yǐngxiǎng.

'Experts **say** that the magnitude and depth of the earthquake will determine its impact on the local community.'

Only sentences without negation markers preceding events were considered for analysis, as including sentences with negation markers could complicate the interpretation of readers' judgments. For instance, readers might naturally doubt that an event occurred, even when it is not explicitly negated (de Marneffe, Manning & Potts 2012). Additionally, the number of such sentences was insufficient for conducting a statistical analysis.

The Chinese evidential hierarchy proposed by Ma (2011) was used to categorize the evidential markers into various types of knowledge in a hierarchical order. The definitions of evidentiality in the hierarchy follow the notion of Chafe (1986), where the speaker's attitude toward the knowledge is considered.

In a news sentence, the speaker's attitude toward the knowledge can be expressed by how they intend for readers to interpret it (Weber 1986). Therefore, for the purpose of collecting readers' judgments regarding the knowledge, anno-

1. The evidential markers in the examples are presented in boldface, while the predicates encoding event knowledge are underlined in this study.

tators were asked to categorize evidential markers based on what they judged the speaker intended in context.

As evidential markers are context-sensitive and polysemous, we recruited two annotators, both graduate students majoring in linguistics and provided them with training sessions to guide them in grouping evidentials into corresponding categories based on context. For example, the evidential marker 澄清 *chéngqīng* ‘clarify’ can convey knowledge in discourse with at least two meanings: as an inferential evidential or as a quotative evidential.

(3a) demonstrates an instance of 澄清 *chéngqīng* ‘clarify’ used inferentially, where the statement made by the store is supported by explicit surveillance footage as evidence. In contrast, (3b) illustrates a quotative usage of the evidential marker, where the source provider (e.g., E.SUN Bank) informs the press about the merger with the shipping company without providing additional explicit evidence. Moreover, instances where evidential markers are employed to convey knowledge based on assumption, speculation, or deduction were excluded from this study (e.g., [4]) as they did not align with any of the evidential categories outlined in the scheme. Annotators were instructed to assign an “other” label to such cases. The inter-rater agreement measured by Cohen’s Kappa (Cohen 1960) was 0.69, reaching the substantial level (Landis & Koch 1977). For cases that did not reach an agreement, the author made the final decision. Details of the annotation results are presented in Section 4.

- (3) a. 店家釋出監視器畫面，澄清自家店員並未加入群毆案件。

Diànjiā shìchū jiānshìqì huàmiàn, **chéngqīng** zìjiā diànyuán bìng wèi jiārù qúnōu ànjàn.

‘The store released surveillance footage and **clarified** that their employees were not involved in the group assault case.’

- b. 玉山銀行於記者會上澄清吞併海運企業一案。

Yùshān Yínháng yú jìzhěhuì shàng **chéngqīng** tūnbìng hǎiyùn qìyè yī àn.

‘E.SUN Bank **clarified** the details of the merger with the shipping company at the press conference.’

- (4) 警方猜測嫌犯持槍逃逸。

Jǐngfāng cāicè xiánfàn chí qiāng táoyì.

‘The police **speculate** that the suspect escaped with a gun.’

3.2 Collection of readers’ judgments

In this study, the design of the questionnaire to collect readers’ judgments, including the presentation of sentences and the inclusion of filter sentences, adhered

to the methodologies established in the studies of de Marneffe, Manning & Potts (2012) and Chang & Hsieh (2020). Each item presented in the questionnaire included an original news sentence, a normalized version, and a five-point scale. The participants were instructed to read a news sentence highlighted with a predicate encoding event knowledge (e.g., 婚約 *hūnyuē* ‘engagement’) and then make a judgment based on the normalized event sentence (e.g., 二人有婚約 *èr rén yǒu hūnyuē* ‘The two had an engagement’) using the five-point scale, depicted in Figure 1. The normalized event sentence, as defined in Saurí (2008) and de Marneffe, Manning & Potts (2012), was designed to help the participants focus on the highlighted event. Readers’ judgments regarding the certainty of events were collected using a five-point Likert scale. The five-point scale consists of the following descriptors: high possibility that it happened (H+), low possibility that it happened (L+), high possibility that it did not happen (H–), low possibility that it did not happen (L–), and unknown (Uu), as shown in Figure 1. The Uu label indicates that readers have no particular knowledge about the probability that an event happened.

高等法院傳訊多位二人好友，均證稱二人有婚約	The High Court summoned several friends of the two individuals, all of whom testified that the two had an engagement.
根據您閱讀後的理解，您是否認為二人有婚約	Based on your reading, do you think that The two had an engagement.
☐ 有一高可能性（已發生/正在發生/未來會發生）	☐ high possibility that it (happened/is happening/will happen)
☐ 有一低可能性	☐ low possibility that it
☐ 無一高可能性（已發生/正在發生/未來會發生）	☐ high possibility that it did not (happened/is happening/will happen)
☐ 無一低可能性	☐ low possibility that it did not
☐ 不知道（無從判斷此事件真實性）	☐ unknown (no daims about the event)

Figure 1. An item in the questionnaire displayed to the participants of this task (only the Chinese version was shown to the participants)

A total of five filter sentences from a previous study (Chang & Hsieh 2020) were included in the questionnaire to ensure participants’ attentiveness to the task. If a participant did not answer one of the filter sentences correctly, all their judgments were discarded.

The Qualtrics platform was used to administer the questionnaire to each participant with the items shuffled.² Each participant was assigned 49 randomly selected items from the dataset, including the five filter sentences, to prevent fatigue while completing the task. Each item may receive different judgments from different participants, as judgments are entirely based on the intuition. Thus, to avoid having each item judged solely by a single participant, each item received eight judgments to reflect the general tendency of the readers. As shown in (5), seven participants regarded the event as certainly having occurred (H+ label), while one participant considered it possibly not having occurred. This result shows a general tendency where the majority of participants believed the event was highly likely to have occurred.

- (5) 根據公平會統計資料顯示，台灣約有 214 萬人、約 9% 人口投身直銷產業。

Gēnjù Gōngpíng huì tǒngjì zīliào **xiǎnshì**, Táiwān yuē yǒu 214 wàn rén, yuē 9% rén kǒu tóushēn zhíxiāo chǎnyè.

‘Statistics from the Fair Trade Commission **reveal** that approximately 2.14 million people, or about 9% of the population, are engaged in the direct selling industry in Taiwan.’

Normalization:

台灣約有 214 萬人、約 9% 人口投身直銷產業。

Táiwān yuē yǒu 214 wàn rén, yuē 9% rén kǒu tóushēn zhíxiāo chǎnyè.

‘Approximately 2.14 million people, or about 9% of the population, were engaged in the direct selling industry in Taiwan.’

Annotations: H+:7, L-:1

The participants were recruited via crowdsourcing from a university communication social group on Facebook in Taiwan, primarily consisting of university students who frequently posted and shared up-to-date information. A total of 357 individuals (217 females, 137 males, and 3 others), all native Mandarin Chinese speakers, were recruited. The average age of the participants was 26 years ($SD=10.2$).

3.3 Procedure for data analysis

Each evidential category contained news sentences annotated with the corresponding evidential markers, as well as the readers’ judgments of those sentences.

2. Qualtrics is an advanced online platform designed for the creation of questionnaires and the management of surveys.

Specifically, if an evidential category included 20 news sentences with associated evidential markers, there would be a total of 160 accompanying readers’ judgments for those sentences. To handle the Likert-type scale data collected from the questionnaire, which included varying evidential categories, an ordinal logistic regression model was suggested (Baayen 2008). Accordingly, an ordinal logistic regression model was employed to assess whether the knowledge encoded in the evidential categories was perceived hierarchically based on readers’ certainty levels, with an alpha level set at $p < .05$.

4. Results

The occurrences of evidential markers in each category are presented in Table 1. It reveals that quotative evidentials account for the largest proportion (77.8%). The function of quotative evidentials is to disclaim authority by the speaker, marking the existence of a specific person or group in the event (Babel 2009, Korta & Zubeldia 2014). This mechanism helps journalists report news objectively and aligns with journalistic protocol (White 2004). Thus, quotative evidentials can be widely observed in the dataset. In contrast, visual and auditory evidentials exhibit the fewest occurrences (3.1% and 0%, respectively). This indicates that news events are less frequently reported with these direct evidentials, as they exhibit a high level of subjectivity (Lyons 1977, Nuyts 2001), which contradicts journalistic protocol. These evidentials are typically used to convey a speaker’s personal experience and imply the speaker’s full responsibility for the reported events (Hsieh 2008). As a result, visual and auditory evidentials are infrequently used in news reporting.

Table 1. The occurrences of evidential markers in the dataset grouped into the five hierarchical evidential categories

Occurrences		Percentage
Visual	21	3.1%
Auditory	0	0%
Inference	85	12.6%
Quotative	523	77.8%
Hearsay	44	6.5%
Total	673	100.0%

The percentage distribution of the hierarchical evidential categories with varying certainty levels is depicted in Figure 2. Most of the judgments lean to the right, signaling that readers tend to make positive judgments about events. The H+ level constitutes the largest proportion (ranging from 57% to 80%) across all the evidential categories, followed by the L+ level (ranging from 9% to 16%) and the Uu level (ranging from 6% to 19%). Only small proportions are given to negative judgments. This indicates that most readers are inclined to believe that a news event is likely to have happened regardless of the type of knowledge introduced. This may be a characteristic of news reporting, where readers are predisposed to believe. It also indicates the existence of mutual knowledge with held beliefs encoded in evidential categories that are shared among conversational participants (Hintz & Hintz 2017).

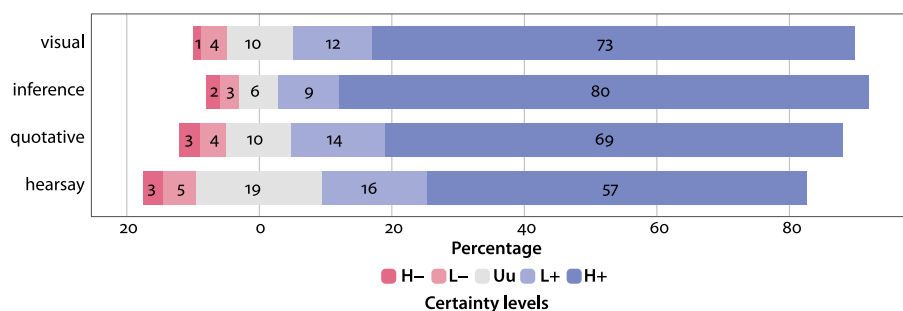


Figure 2. The percentage distribution of the hierarchical evidential categories with varying certainty levels

All evidential categories with collected reader judgments were considered for the ordinal logistic regression model, except for auditory evidentials, which had zero observations. As demonstrated by Hosmer, Lemeshow & Sturdivant (2013), a minimum of 20 observations per category is recommended for the model. The statistical results are presented in Table 2, with a D_{xy} value of 0.081, indicating that the model may reflect random prediction. To further assess the model's performance, a goodness-of-fit test using bootstrapping, as recommended by Baayen (2008), was conducted with 1000 iterations. The bootstrapping results suggest that the model may be slightly overfitted, as the original D_{xy} value of 0.081 was adjusted to 0.080. The small D_{xy} value may also result from the large proportion of readers assigning H+ judgments to news events in this dataset (see Figure 2). Therefore, the model might show a bias toward H+ judgments. However, this bias likely has minimal impact on the overall interpretation of the model, as H+ judgments represent the largest proportion of judgments across all evidential categories in this dataset.

From the statistical results, distinctions were found between the visual and hearsay ($\chi^2[1, N=520]=11.39, p<.05$), inference and quotative ($\chi^2[1, N=4864]=27.96, p<.05$), inference and hearsay ($\chi^2[1, N=1032]=52.81, p<.05$), and quotative and hearsay ($\chi^2[1, N=4536]=21.46, p<.05$) categories, while no significant distinctions were observed between the visual and inference ($\chi^2[1, N=848]=3.42, p=.06$) and visual and quotative ($\chi^2[1, N=4352]=0.90, p=.34$) categories. This indicates that the readers perceived the knowledge encoded in the inference, quotative, and hearsay categories with varying levels of certainty. In contrast, for events expressed via visual evidentials, the readers assigned similar degrees of certainty as they did for events expressed with inference and quotative evidentials.

Table 2. Pairwise statistical results of the ordinal logistic regression model among the evidential categories

	Visual	Inference	Quotative	Hearsay
Visual				
Inference	.06			
Quotative	.34	<.05		
Hearsay	<.05	<.05	<.05	

Additionally, considering the proportion of positive judgments (i.e., H+ and L+) shown in Figure 2, we can infer a hierarchical order among the inference, quotative, and hearsay categories: inference > quotative > hearsay. Although the visual category does not show significant distributional differences compared to the inference and quotative categories in terms of the readers’ judgments, it does exhibit significantly higher proportions of positive judgments compared to the hearsay category: visual > hearsay. Notably, although it ranks the lowest among all categories, the hearsay category exhibits the highest proportions of L+ and Uu judgments. This suggests that the readers were more likely to assign a weak certainty level or remain undecided about events introduced by hearsay evidentials.

Overall, the findings regarding the hierarchical order of evidential categories partially coincide with the hierarchy proposed by Ma (2011), as the proposed hierarchy branches out into two hierarchies based on the observed results. The potential factors contributing to these hierarchical rankings in this study will be discussed in detail in Section 5.

5. Discussion

5.1 The visual category

The utilization of visual evidentials in news reports empowers journalists to organize and present messages from a witness perspective, signifying direct responsibility for the reported event and demonstrating a heightened sense of subjectivity (Hsieh 2008). By conveying a witness viewpoint, journalists tend to provide more detailed depictions of event backgrounds (e.g., time and location), leading to heightened reader involvement and certainty levels (Cornillie 2018). This is shown in (6), which receives 7 H+ judgments and only 1 L+ judgment. The background knowledge enables readers to determine whether the information is verifiable through sight (cf. Musi & Rocci 2017). The more detailed the depiction of a scene, the more salient the speaker's presence in the event, indicating a high degree of subjective commitment to the belief that the event has occurred (Hsieh 2008). This suggests that the use of visual evidentials not only conveys a strong degree of speaker commitment to believing an event but also encourages readers to share a similar level of certainty.

- (6) 台南市麻豆警方昨天深夜在下營區 1 處休閒柚園裡卡拉 OK，查獲越南籍男女辦毒品轟趴。

Táinánshì Mádòu jǐngfāng zuótiān shēnyè zài Xiàying Qū 1 chù xiūxián yòuyuán lǐ kǎlā OK, **cháhuò** Yuènnánjí nǎnnǚ **bàn** dúpǐn hōngpā.

'The MaDou Police Department in Tainan City raided a karaoke bar at a leisure pomelo orchard in Xiaying District last night, **uncovering** a drug party **organized** by Vietnamese men and women.'

Normalization:

越南籍男女辦毒品轟趴。

Yuènnánjí nǎnnǚ bàn dúpǐn hōngpā.

'Vietnamese men and women organized a drug party.'

Annotations: H+:7, L+:1

The readers' degree of certainty towards events introduced by visual evidentials is influenced by both the number of source providers witnessing the event and their identity. When the source provider is a governmental group or community (e.g., 麻豆警方 *Mádòu jǐngfāng* 'the Mádòu Police Department' as shown in [6]), the event sentence receives a higher degree of certainty from readers. In contrast, when the source provider is a random individual (e.g., 楊姓網友 *Yáng xìng wǎngyǒu* 'a netizen with the surname Yang' as shown in [7a]), readers' certainty is comparatively lower. If the individual source provider is associated with a specific

religion (e.g., 引魂師父 *yǐnhún shīfù* 'soul-calling master'), readers' certainty may be even lower, as shown in (7b).

- (7) a. 楊姓網友在臉書社團《爆料公社》發文指控，昨天下午行經國道三號北上出中寮隧道在內側，¹輛銀色三菱轎車行駛在他前面，突然看到該車後座的小孩開啟車窗，頭伸出窗外嘔吐，當下車速還突然轉慢。

Yáng xìng wǎngyǒu zài liǎnshū shètuán 《Bàoliào Gōngshè》 fāwén zhǐkòng, zuótiān xiàwǔ xíngjīng Guódào sān hào běishàng chū Zhōngliáo suìdào zài nèicè, ¹ liàng yínsè Sānlíng jiàochē xíngshǐ zài tā qiánmiàn, tūrán kàndào gāi chē hòuzuò de xiǎohái kāiqǐ chēchuāng, tóu shēnchū chuāng wài ǒutù, dāngxià chēsù hái tūrán zhuǎn màn.

'A netizen with the surname Yang posted in the Facebook group "Breaking News Commune," said that yesterday afternoon while traveling north on National Highway 3 and exiting the Zhongliáo Tunnel, a silver Mitsubishi sedan was driving in front of him. He suddenly saw a child in the back seat of the car open the window, lean out to vomit, and the car immediately decelerate.'

Normalization:

該車後座的小孩開啟車窗。

Gāi chē hòuzuò de xiǎohái kāiqǐ chēchuāng.

'A child in the back seat of the car opened the window.'

Annotations: H+:5, L+:1, Uu:1, L-:1

- b. 據一名有 30 多年經驗的引魂師父透露，檢視頭顱發現有明顯的一處傷口。

Jù yī míng yǒu 30 duō nián jīngyàn de yǐnhún shīfù tòulù, jiǎnshì tóulú fāxiàn yǒu míngxiǎn de yī chù shāngkǒu.

'According to a soul-calling master with over 30 years of experience, an examination of the skull **revealed** a significant wound.'

Normalization:

頭顱有明顯的一處傷口。

Tóulú yǒu míngxiǎn de yī chù shāngkǒu.

'The skull had a noticeable wound.'

Annotations: H+:4, L+:3, Uu:1

5.2 The inference category

Events presented with inference evidentials tend to cause readers to have a high degree of certainty (Cornillie 2018). This may be due to the fact that the use of

inference evidentials presupposes the presence of a set of premises (Musi 2016). Additionally, when the premise is textually expressed, it is easier for readers to reconstruct the speaker's inferential path (Musi & Rocci 2017) and envision the scene of the event (Nuyts 2017, Jaakola 2021), which results in evoking a high level of certainty. The premise is often presented as supplementary evidence in sentences introduced by inference evidentials. For example, in (8), readers believe that the event 車位遭塗銷 *chēwèi zāo túxiāo* 'the parking space had been painted over' introduced by the inference evidential 確認 *quèrèn* 'confirm' certainly happened (receiving 8 H+ judgments) based on the supplementary evidence 該局同仁今上午前往現場勘查 *gāi jú tóngrén jīn shàngwǔ qiánwǎng xiànchǎng kānchá* 'colleagues from the department went to the scene this morning to conduct an investigation.'

- (8) 台中市交通局長王義川證實，該局同仁今上午前往現場勘查，確認車位已遭塗銷。

Táizhōngshì Jiāotōngjúzhǎng Wángyìchuān zhèngshí, gāi jú tóngrén jīn shàngwǔ qiánwǎng xiànchǎng kānchá, **quèrèn chēwèi yǐ zāo túxiāo.**

'Taichung City Traffic Department Director Yi-Chuan Wang confirmed that colleagues from the department went to the scene this morning to conduct an investigation and **confirmed** that the parking space had been painted illegally.'

Normalization:

車位遭塗銷。

Chēwèi zāo túxiāo.

'The parking space had been painted over.'

Annotations: H+:8

To make a premise easily accessible to readers, journalists often provide information with two characteristics: (a) easily recoverable and (b) intersubjectivity accessible (Freeman 2005). Hence, it is observed that the premises presented in inference sentences are often correlated with past events (cf. Fleck 2007, San Roque & Loughnane 2012), as shown in (8). Past events allow readers not only to know that an event happened but also to reconstruct the scene in their minds based on the given facts.

While events expressed by inference evidentials generally evoked strong levels of certainty, there were cases in which a weaker certainty level was provoked. We identified two situations where readers assigned a weaker commitment to the expressed events, which violate the premise of being recoverable and accessible to readers. Firstly, when the premise was based on a poll result, the event received a relatively lower degree of certainty as readers are aware that it is hard to repeat the same poll results, as shown in (9a). Secondly, when the evidence is the result

of research, a lower degree of certainty was assigned to the event, as shown in (9b). It is generally known that research results are preliminary, and it is possible that the same results may not be replicable. Thus, the event could only be partially believed by some people, receiving 5 H+ and 3 H- judgments.

- (9) a. 近郝人士則**指出**，根據黨內一份內部民調，郝龍斌民調排名已經躍居第二，超越吳敦義選情。

Jìn Hǎo rénshì zé **zhǐchū**, gēnjù dǎngnèi yī fēn nèibù míndiào, Hǎolóngbīn míndiào páimíng yǐjīng **yuèjū** dì èr, chāoyuè Wúdūnyì xuǎnqíng.

'The supporters of Mr. Hau's candidacy **indicate** that, based on an internal party poll, Mr. Lung-Bin Hau's poll ranking has **jumped** to second place, diminishing Mr. Den-Yih Wu's electoral prospects.'

Normalization:

郝龍斌民調排名躍居第二。

Hǎolóngbīn míndiào páimíng yǐjīng **yuèjū** dì èr.

'Mr. Lung-Bin Hau's poll ranking has jumped to second place.'

Annotations: H+:4, L+:2, Uu:2

- b. 這件神奇的抗寒外套，使用和 NASA 太空衣一樣的材質氣凝膠，實驗**發現**它能夠抵擋攝氏零下 196 度的低溫，同時內裡維持攝氏 31 度。

Zhèjiàn shénqí de kànghán wàitào, shǐyòng hàn NASA tàikōngyī yīyàng de cáizhí qìníngjiāo, shíyàn **fāxiàn** tā nénggòu **dǐdǎng** shèshì língxià 196 dù de dīwēn, tóngshí nèilǐ wéichí shèshì 31 dù.

'This amazing cold-resistant jacket, made with the same material as NASA space suits—a material called aerogel—was experimented with and **found** to **withstand** temperatures as low as minus 196 degrees Celsius while maintaining an interior temperature of 31 degrees Celsius.'

Normalization:

神奇的抗寒外套能夠抵擋攝氏零下196度的低溫。

Shénqí de kànghán wàitào nénggòu **dǐdǎng** shèshì língxià 196 dù de dīwēn.

'The amazing cold-resistant jacket can withstand temperatures as low as minus 196 degrees Celsius.'

Annotations: H+:5, H-:3

5.3 The quotative category

Quotative evidentials can be utilized to specify that an event is reported by a source provider rather than a journalist, allowing the journalist to distance them-

selves from the source provider (Aikhenvald 2004, Babel 2009, Korta & Zubeldia 2014), and to make readers aware of the existence of a third party other than the involved source provider in an event sentence (Travis 2006). This mechanism allows journalists to disclaim their authority and avoid taking responsibility for the validity of information (Mushin 2013). Thus, the use of quotative evidentials enables journalists to report news objectively without adding their own personal opinions or commitments. Previous studies also concur that quotative evidentials only present the proposition of an event and do not express the speaker's degree of certainty about it (Korta & Zubeldia 2014, Déchaine et al. 2017).

This phenomenon indicates that the assessment of an event's occurrence expressed through quotative evidentials depends on a reader's personal judgment (cf. Mushin 2013). It is observed that when an event introduced by a quotative evidential is reported by a legitimate group of source providers or a legitimate organization, higher levels of certainty are assigned. For instance, in (10a) and (10b), the source provider is a governmental party (i.e., 警方 *jǐngfāng* 'the police') and a legitimate organization (i.e., 成大學生會 *Chéngdà xuéshēnghuì* 'the student union of National Cheng Kung University'), both receiving 8 H+ judgments. These indicators suggest that when an event involves more than one authoritative individual as the source provider, a high degree of certainty is triggered in readers. As journalists distance themselves from reported events via quotative evidentials, readers require more information to determine whether the event is believable. In essence, to decide whether to believe an event happened, readers evaluate whether the knowledge is shared by multiple authoritative individuals.

- (10) a. 警方說，昨天下午約 2 點接獲民眾報案。

Jǐngfāng shuō, zuótiān xiàwǔ yuē 2 diǎn jiēhuò mínzhòng bàozhào.

'The police **said** they received a report from the public yesterday afternoon around 2 o'clock.'

Normalization:

民眾報案。

Mínzhòng bàozhào.

'A report was received from the public.'

Annotations: H+:8

- b. 成大學生會**指出**，南市府日前拆除孫文銅像基座。

Chéngdà xuéshēnghuì **zhǐchū**, Nánshìfǔ rìqián **chāichú** Sūnwén tóngxiàng jīzuò.

'The student union of National Cheng Kung University **pointed out** that the Tainan City Government recently **removed** the pedestal of the Sun Yat-sen bronze statue.'

Normalization:

南市府日前拆除孫文銅像基座。

Nánshìfǔ rìqián chāichú Sūnwén tóngxiàng jīzuò.

'The Tainan City Government recently removed the pedestal of the Sun Yat-sen bronze statue.'

Annotations: H+:8

In contrast, when an event is reported by only one source provider, regardless of whether the source provider is an authoritative politician or their identity is clearly specified, readers tend to exhibit lower degrees of certainty, as shown in (11a) and (11b). If a reported event aligns with a reader's political standpoint or relates to their personal societal beliefs, the event will likely receive a higher certainty judgment, and vice versa. Unlike inference evidentials, quotative evidentials do not provide a premise for readers to infer from. Instead, they rely heavily on their personal knowledge, leading to greater variability in certainty levels compared to the inference category.

- (11) a. 耿爽說，中國會採取一切必要措施保護中國安全。

Gěngshuǎng **shuō**, Zhōngguó huì **cǎiqǔ** yīqiè bìyào cuòshī bǎohù Zhōngguó ānquán.

'Geng Shuang **said** that China would **take** all necessary measures to protect Chinese security.'

Normalization:

中國會採取一切必要措施。

Zhōngguó huì cǎiqǔ yīqiè bìyào cuòshī.

'China would take all necessary measures.'

Annotations: H+:4, L+:2, Uu:1, L-:1

- b. 鄭女則回應，交辦工作都有完成。

Zhèng nǚ zé **huíyīng**, jiāobàn gōngzuò dōu yǒu **wánchéng**.

'Ms. Cheng **responded** that all assigned tasks have been **completed**.'

Normalization:

交辦工作都有完成。

Jiāobàn gōngzuò dōu yǒu wánchéng.

'All assigned tasks have been completed.'

Annotations: H+:3, L+:1, Uu:3, L-:1

5.4 The hearsay category

The way journalists employ hearsay evidentials in depicting events is to imply the existence of an external source (a third group of individuals) (Travis 2006). Hearsay evidentials are used to suggest that the expressed information is shared by both the journalist and the unspecified source, allowing the journalist to distance themselves from responsibility for the reported events and thus achieve objective reporting (Hsieh 2008). Hsieh (2008) further stated that hearsay evidentials are also employed to describe political and business-related topics to convey an objective perspective. Examples of political and business topics expressed using hearsay evidentials are given in (12a) and (12b). In (12a), the use of hearsay evidentials in a political context involves a public figure (i.e., a former president of Taiwan, 馬英九 *Mǎyīngjiǔ* 'Ma Ying-jeou') and illustrates serious and formal affairs (e.g., meeting with Congress members and think tank scholars) to present facts objectively. As for business topics, as shown in (12b), hearsay evidentials often accompany statistics (e.g., 3億元 *3 yì yuán* '300 million dollars') to assist in conveying facts objectively. Since political and business issues are usually more serious, these contexts readily provoke readers to have a higher degree of commitment (e.g., receiving 8 and 6 H+ judgments, respectively).

- (12) a. 據了解，馬英九在華府停留期間，將以接近 1 天半的時間會晤美國多位友台國會議員及智庫學者。

Jù liǎojiě, Mǎyíngjiǔ zài Huáfǔ tíngliú qíjiān, jiāng yǐ jiējìn 1 tiān bàn de shíjiān huìwù Měiguó duōwèi yǒutái Guóhuì yìyuán jí zhìkù xuézhě.

'It is **understood** that during Ma Ying-jeou's stay in Washington, he will spend nearly a day and a half meeting with several pro-Taiwan members of Congress and think tank scholars.'

Normalization:

馬英九將會晤美國多位友台國會議員及智庫學者。

Mǎyíngjiǔ jiāng huìwù Měiguó duōwèi yǒutái Guóhuì yìyuán jí zhìkù xuézhě.

'Ma Ying-jeou will meet with several pro-Taiwan members of Congress and think tank scholars.'

Annotations: H+:8

- b. 現在卻傳出因為獎金預算遭刪 3億元，由原先的102億減少至99億元。

Xiànzài què chuánchū yīnwèi jiǎngjīn yùsuàn zāo shān 3 yì yuán, yóu yuánxiān de 102 yì jiǎnshǎo zhì 99 yì yuán.

'Now it has been **reported** that due to the budget cut of 300 million dollars, the prize budget has been reduced from the original 10.2 billion dollars to 9.9 billion dollars.'

Normalization:

獎金預算遭刪3億元。

Jiǎngjīn yùsuàn zāo shān 3 yì yuán.

'The prize budget has been cut by 300 million dollars.'

Annotations: H+:6, L+:2

Additionally, the implementation of hearsay evidentials in describing events can serve a narrative communicative function (Hsieh 2008). Narration aims to illustrate the scenario of an event's occurrence, encompassing elements such as the participants, time, location, and more, as exemplified in (13a) and (13b). Utilizing hearsay evidentials with a narrative function can provide an unbiased account of a reported event, enabling readers to perceive the event as if they were present and apply their own social norms to make judgments (Hsieh 2008). Thus, when a hearsay sentence provides more detailed narrative depictions of an event, it has a stronger effect in involving the readers, thereby triggering higher confidence in believing the reported event. Conversely, when the narrative depiction is less detailed, readers have weaker involvement in the story, leading to a lower degree

of commitment. For example, comparing (13a) and (13b), it is observed that (13a) provides more detailed scenario descriptions of an event and received higher certainty levels, including where Suhua Han originated from, what she did in Hong Kong, and her identity in Hong Kong. On the contrary, (13b) only specifies a fact that a woman was found dead at a location, without providing further detailed information explaining how or why this happened, and thus received lower certainty levels from the readers.

- (13) a. 據了解，韓素華原籍上海，後到香港經商，並取得香港居民身分。

Jù liǎojiě, Hánsùhuá yuánjí Shànghǎi, hòu dào Xiānggǎng jīngshāng, bìng qǔdé Xiānggǎng jūmín shēnfēn.

‘It has been **learned** that Suhua Han, originally from Shanghai, later moved to Hong Kong for business, and obtained Hong Kong residency.’

Normalization:

韓素華原籍上海。

Hánsùhuá yuánjí Shànghǎi.

‘Suhua Han is originally from Shanghai.’

Annotations: H+:7, L-:1

- b. 台灣一名婦人驚傳陳屍在泰國芭達雅一家飯店的泳池。

Táiwān yī míng fùrén jīng chuán chénshī zài Tàiguó Bādáyǎ yī jiā fàndiàn de yǒngchí.

‘A Taiwanese woman was **reported** to be found dead in a swimming pool at a hotel in Pattaya, Thailand.’

Normalization:

婦人陳屍在泰國芭達雅一家飯店的泳池。

Fùrén chénshī zài Tàiguó Bādáyǎ yī jiā fàndiàn de yǒngchí.

‘A woman was found dead in a swimming pool at a hotel in Pattaya, Thailand.’

Annotations: H+:3, L+:2, Uu:3

Since the use of hearsay evidentials does not involve a journalist’s degree of certainty to an event, it is possible that readers may exhibit a lower degree of certainty (Cornillie 2018). In this study, the hearsay category has the lowest certainty level compared to other evidential categories (see Figure 2). This can be attributed to the fact that most hearsay evidentials provide minimal scenario details, offering insufficient reasoning for readers to mentally reconstruct the scene. This is understandable, as journalists typically use hearsay evidentials when they lack full information of a reported event. Consequently, journalists are less likely to

detail the event, instead presenting a simple description of the event, which leads readers to have lower confidence and assign more L+ and Uu judgments.

6. Conclusion

This study examines whether the hierarchy of Chinese evidential categories proposed by Ma (2011) maintains the same ranking when the encoded knowledge is judged by readers. Our findings suggest that the proposed hierarchy may need slight modifications when assessed from the perspective of readers' degree of certainty. While knowledge expressed by the visual category receives significantly higher certainty levels from readers than the hearsay category, its judgment distribution is similar to the inference and quotative categories. Additionally, we identified a hierarchical order among the inference, quotative, and hearsay categories, ranking from highest to lowest.

From the discussion in Section 5, we summarize three factors that may influence readers' degree of certainty regarding knowledge of events presented via evidentials: (a) the provision of evidence or detailed event depictions, (b) whether knowledge of an event is given by more than one legitimate source provider or by a legitimate organization, and (c) whether knowledge of an event aligns with readers' political standpoints or personal societal beliefs.

The two evidential hierarchies observed in this study can be further explained as follows. Inference evidentials come with explicit premises that provide sufficient evidence for readers to infer from, thus evoking higher certainty levels than quotative and hearsay evidentials. One major difference between quotative and hearsay evidentials lies in whether the journalist knows the specific source provider of the event in question. Quotative evidentials are more likely to include precise and detailed event information, as journalists can access information by interviewing the source provider, leading to higher certainty levels than hearsay evidentials. Regarding the visual category, it is observed that it presents knowledge from a witness perspective, often involves multiple source providers, and evokes high certainty in readers. These characteristics are also found in the contexts of inference and quotative categories, which elicit high certainty levels. This may explain why no significant differences are found between the visual and inference categories, and between the visual and quotative categories.

There are some limitations of this study, which could be addressed in future research. For example, auditory evidentials were not analyzed in this dataset as they did not occur. Thus, future studies should collect sentences with auditory evidentials for analysis. Although evidential markers that convey assumption, speculation, and deduction are not considered in this hierarchy scheme, they may

exhibit a similar distribution of certainty levels to some categories within the hierarchy. Additionally, sentences containing negation markers were not analyzed in this study. Future research could include this aspect by collecting a sufficient number of sentences with negation markers for comparison.

In general, this study contributes to understanding how knowledge expressed by evidentials is assessed by readers and identifies potential factors affecting readers' varying certainty levels towards event knowledge. The findings can be applied to other disciplines. For instance, the results may encourage the field of journalism to develop more consistent reporting protocols to provide trustworthy information. Additionally, the results could assist the education sector in guiding learners to enhance their media literacy when reading news reports.

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